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Environmental Racism in the Accessibility of Urban Green Space: A Case Study of a Metropolitan Area in an Emerging Economy

Adriano Bressane ^{1,2,*} , Anna Isabel Silva Loureiro ² and Rogério Galante Negri ¹

¹ Institute of Science and Technology, São Paulo State University, São José dos Campos 12245-000, Brazil

² Graduate Program in Civil and Environmental Engineering, São Paulo State University, Bauru 17033-360, Brazil

* Correspondence: adriano.bressane@unesp.br

Abstract: Urban Green Spaces (UGS) are integral to advancing urban sustainability and improving the quality of life in cities. However, in rapidly urbanizing regions like the São Paulo Metropolitan Region (MRSP), significant environmental injustices in UGS accessibility present a complex challenge that requires in-depth understanding. Notably, existing studies predominantly focus on developed countries, leaving a gap in research concerning emerging economies in the Global South. This study aims to analyze the associations between sociodemographic factors and environmental racism in UGS accessibility within the municipalities of MRSP. The research utilizes Spearman Rank Correlation and multiple linear regression analyses on data sourced from the Brazilian Institute of Geography and Statistics and the Urban Green Data Platform. Key variables include the number of inhabitants, territorial area, population density, urbanization rate, gross domestic product (GDP), human development index (HDI), urban vegetation coverage, UGS per capita, and the difference between the total population and the Black or Indigenous populations residing outside the vicinity of UGSs as an indicator of environmental racism. The findings reveal significant correlations between higher GDP and HDI with increased environmental racism in UGS accessibility, suggesting that, in the absence of equitable policies, economic and human development may exacerbate disparities in green space distribution. Moreover, the study demonstrates that increased urban vegetation coverage is significantly associated with reduced environmental disparities, underscoring the role of urban greenery in mitigating inequality. These results emphasize the need for comprehensive urban planning and targeted policies that prioritize the equitable development of UGS, particularly in underserved areas. Future research should explore longitudinal data to establish causality and consider additional variables such as political governance and cultural factors, which could provide a more comprehensive understanding of environmental racism in UGS accessibility.

Keywords: environmental racism; urban green spaces; sociodemographic factors; metropolitan regions; emerging economies



Citation: Bressane, A.; Loureiro, A.I.S.; Negri, R.G. Environmental Racism in the Accessibility of Urban Green Space: A Case Study of a Metropolitan Area in an Emerging Economy. *Urban Sci.* **2024**, *8*, 224. <https://doi.org/10.3390/urbansci8040224>

Academic Editor: Luis Hernández-Callejo

Received: 9 October 2024

Revised: 12 November 2024

Accepted: 19 November 2024

Published: 21 November 2024



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1. Introduction

Urban Green Spaces (UGS), encompassing parks, squares, gardens, and ecological corridors, perform a multifunctional role in city environments by improving air quality, mitigating heat islands, fostering biodiversity, and providing spaces for recreation and social interaction [1–5]. In rapidly urbanizing regions, where the process of “fast urbanization” accelerates urban growth, the equitable accessibility of UGSs becomes even more crucial to ensure that all urban residents can fully benefit from these essential green amenities. Overall, fast urbanization refers to the accelerated expansion of urban areas, which often outpaces the development of essential infrastructure, including transportation networks and adequate housing, resulting in various environmental and social challenges [6,7].

Recent studies have emphasized the role of city morphology in shaping community behavior, particularly regarding how the physical and social environment influences residents' interactions and access to resources. Balsa-Barreiro and Menendez [8] provide a comprehensive review of this topic, exploring how urban form affects social dynamics through factors such as connectivity, accessibility, and environmental quality. The authors discuss how different urban configurations can either facilitate or hinder social cohesion, mobility, and equitable access to amenities like UGSs, underscoring the significance of thoughtful urban planning. This aligns with the current study's findings, highlighting that the spatial distribution of UGSs not only reflects socio-economic disparities but also actively shapes patterns of social behavior and environmental justice within metropolitan regions.

However, the distribution of UGSs frequently reflects broader socio-economic disparities, particularly in metropolitan regions, where urban expansion has intensified existing social inequalities and further entrenched patterns of environmental injustice [9]. Han et al. [10] conducted a comparative study on global urban green provision, demonstrating that cities in higher-income regions generally offer more equitable access to green spaces than those in lower-income regions. This finding underscores the systemic inequities that marginalized communities face in accessing UGSs, which often lack adequate maintenance and infrastructure, further contributing to health disparities [11]. Additionally, the social and ecological trade-offs of greening cities must be considered, as efforts to enhance UGSs can inadvertently lead to processes like gentrification, where the benefits of urban renewal may disproportionately favor higher-income groups while exacerbating social exclusion for less affluent residents. These dynamics underscore the need for targeted urban planning policies that address the unequal distribution of green spaces in rapidly changing urban landscapes, promoting equitable access to recreational areas and enhancing residents' quality of life.

Larger cities with high population densities face complex challenges, including competition for limited UGSs and entrenched inequalities in the distribution of these green spaces, often resulting from long-standing social and economic disparities [9]. However, urbanization dynamics are not confined to major metropolitan areas. As highlighted by Balsa-Barreiro et al. [6], these dynamics occur at multiple spatial scales, affecting medium and small cities where the urbanization process may be rapid, inadequate, or unbalanced. In such contexts, the uneven distribution of UGSs can be particularly pronounced, exacerbating social inequities in access to green spaces. In municipalities with extensive territorial areas, UGS resources may still be unevenly distributed [12]. Even in economically prosperous regions with high human development indices, equitable UGS distribution is not guaranteed [13]. Collectively, these conditions suggest that marginalized communities often face limited access to UGSs with adequate maintenance and infrastructure, reinforcing socio-environmental inequalities.

In this context, environmental racism, a specific form of environmental injustice, refers to the disproportionate exposure of marginalized racial and ethnic communities to environmental hazards and the corresponding lack of access to environmental benefits, such as green spaces [14,15]. Brazil [16] extends this understanding by examining environmental inequality across different geographic scales in American cities, revealing that residents from minority and low-income neighborhoods not only face higher pollution levels in their residential areas but also encounter greater environmental burdens in the neighborhoods they travel to for work and leisure. This multi-scale exposure highlights the complexity of environmental racism and reinforces its presence in everyday urban mobility. In many cases, predominantly minority communities are situated in areas with fewer or poorly maintained UGSs, exposing them to greater environmental stressors and fewer opportunities for health and well-being [17]. This inequity is not merely a consequence of economic factors but is also deeply rooted in racial discrimination, which influences how urban resources, including UGSs, are allocated and maintained [18].

Kabisch and Haase [19] examined the provision of UGSs in Berlin, Germany, and found that areas with higher socio-economic status enjoyed a more equitable distribution

of green spaces, underscoring the necessity for policies addressing environmental racism and socio-environmental disparities to ensure equitable access to UGSs. Similarly, Ekkel and de Vries [20] evaluated the relationship between nearby green space and human health, emphasizing the inequalities in exposure to the benefits of UGSs, which disproportionately affect marginalized communities.

Jennings et al. [21] explored the cultural ecosystem services provided by UGSs, focusing on equity and social determinants of health, further highlighting that environmental racism often intersects with public health disparities. The study stresses that addressing inequities in green space distribution is essential for improving the well-being of underrepresented racial and ethnic groups. Maas et al. [22] conducted a study in the Netherlands, discovering that communities with higher human development indices had better access to green spaces. They concluded that improving human development can reduce inequities in UGS distribution and contribute to better health outcomes.

In South America, environmental inequities in UGS provision have been widely documented. In Argentina, Fernandez et al. [23] found significant disparities in green space availability in Buenos Aires, where affluent neighborhoods had better-maintained parks and higher per capita green space compared to marginalized areas. Similarly, Ribeiro et al. [24] reported that in Brazil's metropolitan regions, the spatial distribution of UGSs is heavily skewed towards wealthier and predominantly White areas, leaving Afro-Brazilian communities underserved. Furthermore, Colombia's capital, Bogotá, has been a focal point for studies demonstrating that socioeconomic status strongly influences access to recreational parks, with low-income communities experiencing severe deficits in green space provision [9].

In a broader context of socio-environmental disparities, the inverse relationship between population density and green space availability, as observed in both Europe and Asia, highlights the role of urbanization patterns in shaping green space accessibility. Asian cities have experienced rapid and dense urban growth, particularly in the latter half of the 20th century, driven by accelerated industrialization and population migration to urban centers. This has resulted in higher population densities and fewer green spaces per capita, as urban expansion often prioritizes economic development over environmental considerations. In contrast, the growth of European cities has historically been more organized and sprawling, facilitated by stricter urban planning regulations and policies aimed at integrating green spaces within urban areas [23]. Despite these efforts, the inverse relationship in Europe is surprising and may reflect the ongoing challenges in balancing densification with the restoration or preservation of green spaces. As cities continue to pursue densification strategies to accommodate growing populations, green space recovery efforts may struggle to keep pace, potentially explaining the persistent disparities.

Based on these different developmental trajectories, a hypothesis can be proposed: while densification in Asian cities has primarily resulted from rapid urbanization, leading to limited space for UGSs, European cities may face a legacy of industrial-era land use patterns that constrain the expansion of green spaces in already built-up areas. The recent initiatives to restore UGSs in Europe indicate a shift towards more sustainable urban planning, reflecting a growing commitment to environmental and social well-being. However, these efforts might not be sufficient to counterbalance the historical loss of green areas in densely populated neighborhoods.

A correlation between environmental inequalities and economic development indicators has been highlighted, with Pulido [18] and Pellow [25] both arguing that communities with higher economic and human development levels can exert more political power to secure environmental benefits, often at the expense of marginalized communities. These inequalities are further explored by Mohai et al. [26], who suggest that urban policies must consider these socio-economic factors to mitigate environmental racism.

The need for continuous monitoring and data-driven decision-making in urban growth and green space policies is emphasized by Haase et al. [11] and McPhearson et al. [27]. Both studies advocate for the integration of socio-economic and environmental data in

urban planning, highlighting that robust monitoring systems are crucial for evaluating the effectiveness of UGS policies and ensuring sustainable urban development.

Despite the well-documented environmental inequalities in UGS accessibility, current research predominantly focuses on developed countries, leaving a gap in studies on emerging economies, where urbanization dynamics and socio-economic challenges differ markedly [1,24,28–31]. The unique socio-economic diversity and rapid urban growth of the São Paulo Metropolitan Region (MRSP), Brazil, present an ideal case study to address this gap. Therefore, this study aims to examine the associations between sociodemographic factors and environmental racism in UGS accessibility within the MRSP, providing empirical evidence to inform targeted urban policies.

2. Material and Method

2.1. Study Area

The research area encompasses 39 municipalities within the MRSP, a region that stands as one of the largest urban conglomerates in the world, with a population exceeding 21 million inhabitants. The MRSP plays a crucial role in Brazil's economy, accounting for approximately 18% of the country's Gross Domestic Product (GDP). Characterized by rapid urbanization and diverse socioeconomic conditions, the region faces significant environmental challenges, particularly concerning the equitable distribution and management of UGSs [32–34].

The demographic dynamics of Latin America and the Caribbean, where São Paulo is a major urban hub, are marked by one of the highest levels of urbanization globally, with over 80% of the population residing in urban areas [35]. This extensive urbanization is driven by a demographic transition that has accelerated since the mid-20th century, characterized by declining birth rates and increasing life expectancy. These shifts have influenced social structures and intensified the demand for urban infrastructure, including green spaces, while exacerbating socioeconomic disparities.

In this context, São Paulo exemplifies the broader trends observed in the region, where cities have expanded rapidly, often without adequate urban planning. The resulting uneven distribution of UGSs reflects not only the challenges of managing high population density and competing land use demands but also the deep-seated social inequalities that persist in the urban fabric [36,37]. Recent studies have highlighted that while economic growth has significantly reshaped urban areas, it has not necessarily translated into equitable access to environmental amenities [38,39]. This underscores the importance of targeted urban policies that prioritize the integration of green spaces as essential components of sustainable development and social equity.

The pressures on UGSs in the MRSP stem from rapid urban growth, high population density, and the intense competition for land use, making their management a critical issue. These challenges are further complicated by socio-environmental disparities that limit marginalized communities' access to well-maintained UGSs. Addressing these issues requires a comprehensive understanding of the region's demographic and socio-economic context to inform policies that promote equitable and sustainable urban development. Such policies must prioritize inclusive access to green spaces, aiming to balance urban expansion with environmental justice and resilience in the face of ongoing urbanization.

2.2. Data Collection

Data on municipalities within the MRSP were collected from the Brazilian Institute of Geography and Statistics (IBGE) [40] and the Urban Green Data Platform [41] to obtain information at the municipal scale regarding the following:

- Municipality Size: this metric comprises three variables:
 - Number of Inhabitants: The total population residing in each municipality, based on the most recent census data.
 - Territory Area (km²): The total land area of the municipality, measured in square kilometers.

- Population Density (inhabitants/km²): The number of people per square kilometer, calculated by dividing the total number of inhabitants by the territory area.
- Urbanization Rate (%): this refers to the percentage of the total municipal area classified as urban land. The classification follows criteria set by the IBGE, which defines urban areas based on the density of built-up infrastructure, population concentration, and the presence of urban services and amenities.
- Gross Domestic Product (GDP): economic output per inhabitant.
- Human Development Index (HDI): a composite index that measures average achievement in the dimensions of health, education, and standard of living.
- Urban Vegetation Coverage (%): the percentage of urban territory covered by vegetation.
- UGS per capita (m²/inhabitant): the amount of green space available per inhabitant.
- Population served by UGS (%): the proportion of the population residing within a 400 m radius of UGS, i.e., in the greenspace vicinity [42].
- Environmental Racism in UGS Accessibility (ERUA): the percentage difference between the total population and the Black or Indigenous populations served by UGS (Equation (1)):

$$ERUA(\%) = \left(1 - \frac{\text{Black or Indigenous populations served by UGS}(\%)}{\text{Total population served by UGS}(\%)} \right) \times 100, \quad (1)$$

Equation (1) was adopted to represent relative inequality, quantifying the extent to which Black or Indigenous populations are underrepresented relative to the total population served by UGSs.

Each of the municipalities in the MRSP was considered an individual sampling unit for data collection ($N = 39$), providing a comprehensive analysis of the entire region. The data collection employed advanced remote sensing techniques through platforms like Google Earth Engine (GEE) for geospatial analysis, utilizing satellite imagery (Sentinel-2 and Landsat 8) to derive vegetation indexes such as Normalized Difference Vegetation Index (NDVI) and the Percentual Vegetation Coverage (PVC), thereby enhancing the accuracy and spatial resolution of the information used in this study.

2.3. Data Analysis

For the analysis, the data collection methodology distinguishes between different types of vegetation. The UrbVerde platform (<https://urbverde.iau.usp.br/#/> (accessed on 11 December 2023)), the source of the data used in the present study, specifically accounts for green areas such as public parks, plazas, and ecological corridors that are accessible to the general population. In this platform, the vegetation data are categorized based on public accessibility using the land use classification from the OpenStreetMap database, combined with criteria for minimum size and urban area designation. Only public green spaces larger than 5000 m² (0.5 hectares) were considered in the analysis of UGS metrics, including “UGS per capita” and “Population served by UGS”.

The Spearman Rank Correlation (ρ) test was utilized due to its suitability for the heterogeneous distribution within the dataset. This test was employed to assess the strength and direction of the linear associations between environmental racism and sociodemographic factors, aiming to quantify the degree of association between these variables. Based on these results, an exploratory analysis was conducted to develop a conceptual diagram, with a focus on identifying interaction effects among the variables. Subsequently, after normalizing the data using Z-score transformation, a multiple linear regression analysis was performed on interaction terms to confirm or reject the interaction effects depicted in the previously constructed conceptual diagram. This methodological approach enabled the exploration and quantification of both direct and indirect association between the variables. All statistical tests were conducted at a significance level of 0.05.

3. Results

Table 1 provides the results of the Spearman correlation analyses from the association between sociodemographic factors and environmental racism in UGS accessibility within the MRSP. The findings reveal that certain variables, such as the number of inhabitants, population density, urbanization rate, GDP, and HDI, exhibit positive correlations with environmental racism. In contrast, urban vegetation coverage and UGS per capita are negatively correlated. The significant correlations indicate that these factors may play a crucial role in managing environmental disparities in UGS accessibility.

Table 1. Sociodemographic factors and environmental racism in UGS accessibility.

	ρ	<i>p</i> -Value
Number of inhabitants	0.63	<0.001
Territorial area	−0.16	0.3625
Population density	0.64	<0.001
Urbanization rate	0.67	<0.001
Gross domestic product (GDP)	0.68	<0.001
Human development index (HDI)	0.65	<0.001
Urban vegetation coverage	−0.75	<0.001
UGS per capita	−0.38	0.0315

The table presents the Spearman correlation coefficients (ρ) and corresponding *p*-values for the associations between sociodemographic factors and environmental racism in UGS accessibility. Correlation strength is interpreted based on Cohen's [43] guidelines: weak (0.10–0.29), moderate (0.30–0.49), and strong (0.50 and above). A *p*-value < 0.05 indicates statistical significance. N = 39.

The multiple linear regression analysis, shown in Table 2, reveals the interacted influence of these sociodemographic factors on environmental racism in UGS accessibility within the MRSP. This analysis provides deeper insights into the collective impact of these variables, identifying significant predictors and their respective confidence intervals.

Table 2. Interaction effects of sociodemographic factors on environmental racism in UGS accessibility.

Interaction	β	<i>p</i> -Value	95% CI	
			Lower	Upper
Constant	−15.85	0.0	−25.94	−5.76
Urbanization Rate * Population Density	13.96	0.05	−0.3	28.22
GDP * Human development index	15.69	0.01	4.89	26.48
Urban Vegetation Coverage * UGS per Capita	−6.42	0.36	−20.41	7.57

N = 39.

Based on the significant correlations, it becomes evident that certain sociodemographic factors not only influence environmental racism individually but may also interact in ways that amplify their interaction effects. This analysis suggests that some variables, particularly those showing moderate to strong correlations with environmental racism, may have interconnected impacts; in particular:

- Urbanization Rate and Population Density: higher urbanization rates may amplify the effects of population density on environmental racism.
- GDP and HDI: economic output could interact with human development indicators, influencing disparities in UGS accessibility.
- Urban Vegetation Coverage and UGS per Capita: the availability of green spaces per inhabitant may be influenced by the overall vegetation coverage in urban areas.

In light of the potential interaction effects, the conceptual diagram in Figure 1 visually represents these relationships, offering a comprehensive framework to understand how sociodemographic factors may collectively shape environmental racism in UGS accessibility.

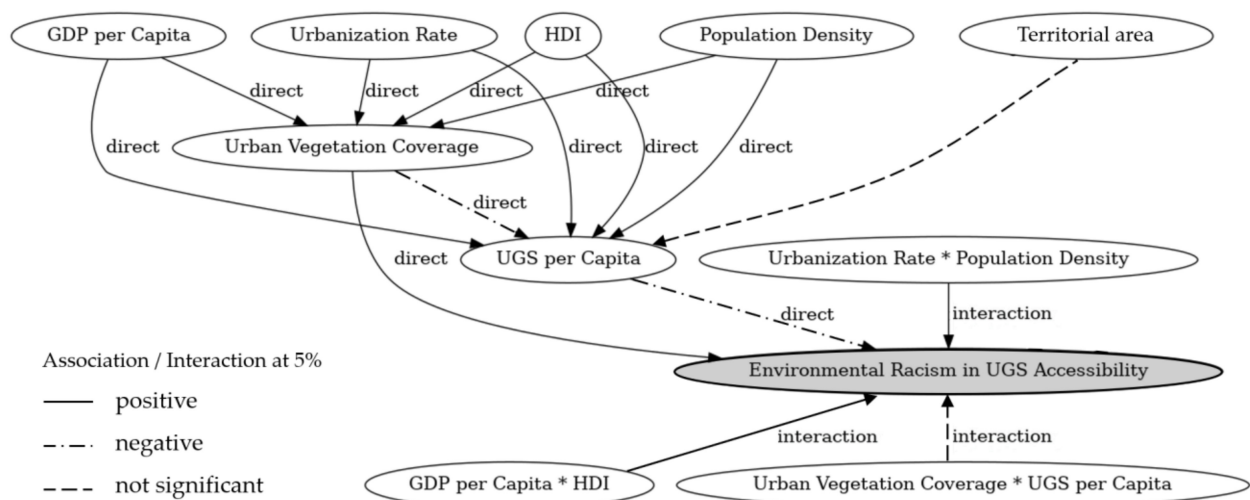


Figure 1. Relationships between sociodemographic factors and environmental racism in urban green space accessibility.

Table 2 presents the significance of the interaction effects analyzed, where the interacted influences of urbanization rate with population density, and GDP with HDI, significantly affect environmental racism in UGS accessibility.

It is worth clarifying that the decision to focus on specific interaction terms in the multiple regression analysis was based on the results of the correlation tests, which identified sociodemographic factors with moderate to strong associations with environmental racism in UGS accessibility. Interaction terms were selected to explore whether pairs of variables with significant individual associations had combined effects on disparities in green space access. Although all interactions depicted in Figure 1 were tested, only those with significant or theoretically relevant results are reported in Table 2 to avoid diluting the interpretations with numerous non-significant findings. This approach allows for a focused discussion on key interaction effects that may inform future research on social and environmental inequalities.

4. Discussion

4.1. Impact of Socioeconomic Development

This study found a significant positive correlation between GDP and environmental racism in UGS accessibility, indicating that higher economic development is associated with increased environmental disparities in UGS allocation. This finding is consistent with a study conducted in Tehran, which found that areas with higher socioeconomic status had better access to UGSs compared to lower socioeconomic areas, thereby reducing environmental injustices [42]. However, our analysis also indicated that some interaction effects initially considered were not statistically significant at the $\alpha = 0.05$ level, as evidenced by the confidence intervals including zero. As a result, we have revised the interpretation to emphasize that while GDP shows a significant direct correlation, other interaction effects require cautious interpretation due to their lack of statistical support. The focus remains on the clear relationships identified, such as the significant interaction between GDP and HDI, which provides a more robust understanding of how economic resources can influence UGS accessibility when coupled with human development indicators.

Furthermore, the negative correlation found between urban vegetation coverage and environmental racism aligns with the broader literature. A study in China highlighted how socioeconomic factors, including high housing prices, influence green space accessibility, with wealthier communities benefiting more from these amenities [30]. This suggests that while socioeconomic development can enhance green space availability, it may also exacerbate inequalities if not coupled with targeted policies aimed at equitable distribution. These

results underscore the complexity of the relationship between socioeconomic development and environmental justice in UGSs. While economic growth has the potential to drive improvements in urban infrastructure and increase the availability of green spaces, it may also exacerbate some forms of environmental racism and reinforce existing disparities if not carefully managed through inclusive urban planning. The findings suggest that socioeconomic development alone may not always lead to equitable access, as systemic inequalities can persist despite overall economic growth.

4.2. Role of Urban Vegetation Coverage

The research revealed a negative correlation between urban vegetation coverage and environmental racism, indicating that increased vegetation is associated with more equitable access to green spaces. In the context of the MRSP, this association can be explained by the concentration of urban vegetation in wealthier areas, where municipal investments in green spaces are higher. This tends to leave underprivileged and predominantly Black or Indigenous communities with less access to well-maintained green spaces, perpetuating environmental inequalities. The study also found that while interactions between urban vegetation coverage and UGS per capita were explored, the statistical analysis showed that these interactions were not significant, highlighting the need to focus on direct correlations rather than overinterpreting non-significant findings. The complexities observed reflect the broader literature's indication that the relationship between urban vegetation coverage and environmental disparities is context-dependent, requiring nuanced urban planning strategies [44].

4.3. Interplay Between Population Density and Urbanization Rate

The findings underscore that both population density and urbanization rate have significant individual associations with the spatial distribution and accessibility of UGS. However, the analysis revealed that the interaction term between these variables was not statistically significant, suggesting that while each factor independently influences green space accessibility, their combined effect did not reach the threshold for statistical significance in this study. This finding aligns with studies indicating that the effects of population density on UGSs may vary depending on other contextual factors, such as urban planning policies or geographic characteristics [45,46]. Therefore, the results support the notion that while urban density and development levels impact green space distribution, interactions between these factors may be more complex and contextually specific than initially expected.

4.4. Interaction Effects of HDI and GDP

The significant interaction between GDP and HDI suggests that socioeconomic development, when coupled with high human development, may lead to better outcomes in terms of UGS accessibility. This finding aligns with previous studies showing that higher levels of economic and social development can increase the availability of UGSs, though the benefits may not always be evenly distributed across all population groups [43]. Our results emphasize that GDP and HDI together contribute to shaping environmental justice outcomes in urban settings, highlighting the dual role of economic factors in both promoting and potentially exacerbating inequalities in UGS accessibility.

5. Recommendations for Enhancing Equitable UGS Accessibility

This study highlights the importance of considering both horizontal and vertical equity in the distribution of UGSs. This aligns with findings from Wuhan, China, where policymakers are urged to address inequities across different social groups and geographic areas, ensuring that green spaces are accessible to all populations [47].

Furthermore, in Portland, Oregon, a methodology was developed to identify UGS inequities by evaluating the distribution of parks and green cover across sociodemographic populations. This approach could serve as a model for other cities aiming to promote

distributional justice in UGS allocation [48]. However, efforts to improve green spaces in underserved neighborhoods may lead to unintended consequences such as green gentrification, where long-term, lower-income residents are displaced due to rising property values and rents [49]. This issue, discussed in the context of environmental justice, underscores the need for policies that prioritize diversity and inclusiveness while expanding green space accessibility [50].

5.1. Prioritize UGS Development in Areas with Low GDP and HDI

This study identified a significant positive correlation between GDP, HDI, and environmental racism in UGS accessibility. To mitigate these disparities, urban planners should prioritize the development and maintenance of UGS in areas with lower GDP and HDI. By focusing on these less affluent regions, planners can ensure that all residents, regardless of their economic status, have equitable access to green spaces. Similar approaches have been advocated in cities like Wuhan, China, where socioeconomic factors were found to significantly influence the spatial equity of green spaces, necessitating targeted interventions in less affluent areas [47]. Another notable example is the participatory design approach in Rosario, Argentina, which integrated community feedback into urban agriculture and green space development to address socio-economic inequities [51]. Such practices demonstrate the potential of collaborative urban planning to improve access in underserved areas.

5.2. Implement Policies to Enhance Vegetation Coverage Without Exacerbating Inequities

The research revealed a negative correlation between urban vegetation coverage and environmental racism, indicating that increased vegetation coverage is associated with more equitable access to green spaces. Urban planners should implement policies that not only increase vegetation coverage but also ensure that this coverage benefits all socioeconomic groups equally. This could involve the strategic placement of well-equipped and well-infrastuctured green spaces in underserved communities, as was demonstrated in Portland, Oregon, where a methodology was developed to identify and rectify UGS inequities across sociodemographic populations [48]. In Caracas, Venezuela, participatory urban management was employed to revitalize neglected green areas, balancing community needs and sustainable development goals [52]. Such initiatives underscore the importance of inclusivity in green space planning.

5.3. Optimize UGS Allocation in High-Density Areas to Balance Population Density and UGS Needs

Given the positive correlation between population density and UGS accessibility, it is crucial to optimize the allocation of green spaces in high-density areas. This might involve innovative urban design solutions such as vertical gardens, rooftop green spaces, and the conversion of underutilized urban areas into public green spaces. In Seoul, South Korea, efforts to ensure equitable access to green spaces for marginalized groups, including those in high-density areas, have shown that careful planning can overcome the challenges posed by dense urban environments [48]. In turn, in Bogotá, Colombia, urban planners collaborated with local communities to introduce rooftop gardens and transform underutilized spaces into community parks, showcasing how innovative designs can address spatial constraints [52].

5.4. Address Urbanization-Driven Inequities in UGS Distribution

This study found that higher urbanization rates are associated with decreased environmental racism in UGS accessibility. To support this trend, urban policies should be designed to manage urbanization in a way that includes equitable green space distribution [53]. In Al-Mughayyer, a town in the Middle East, a similar approach was taken to ensure that urban expansion did not exacerbate existing inequalities in green space accessibility, demonstrating the importance of proactive planning in rapidly urbanizing areas [54]. In São Paulo, Brazil, participatory budgets have been used to prioritize investment UGSs

for marginalized communities, ensuring equitable access while fostering community involvement [55,56]. This approach highlights the potential of democratized decision-making in addressing disparities.

5.5. Incorporate Community Participation in UGS Planning and Maintenance

To ensure that UGS planning reflects the needs of all community members, particularly marginalized groups, it is essential to incorporate community participation in decision-making processes. Mechanisms such as public consultations, participatory design workshops, and community-led maintenance programs can empower residents to have a say in how green spaces are developed and maintained. This approach has been shown to be effective in various urban contexts, where active community involvement has led to better equity in green space distribution [57]. In Quito, Ecuador, community-led green space maintenance programs have empowered local residents, improving both the equity and sustainability of urban greenery [58]. Similarly, participatory workshops in Montevideo, Uruguay, have demonstrated how collaborative design processes can lead to more inclusive UGSs [59].

6. Limitations and Future Research

This study provides significant insights into the interplay between socio-demographic factors and environmental racism in UGS accessibility within the MRSP, but it has some limitations.

A key limitation is the reliance on secondary data, which may not fully capture the nuances of marginalized communities' experiences with UGS accessibility. Future research should incorporate primary data collection, such as household surveys or interviews, to gain more detailed, context-specific insights. Additionally, the cross-sectional nature of the study limits the ability to infer causality. Longitudinal studies are recommended to explore how these relationships evolve over time, particularly as urbanization and socio-economic conditions continue to change within the MRSP.

The study's exploratory approach, utilizing independent correlation tests and multiple linear regression, has limitations that warrant consideration. While the statistical methods applied helped identify associations and interaction effects, they may not fully capture the complexity of relationships between sociodemographic factors and environmental racism in UGS accessibility. Path analysis or structural equation modeling (SEM) could be employed in future research to model conditional dependencies and causal pathways more accurately, particularly with a well-established theoretical framework and longitudinal data. Additionally, refining the treatment of interaction terms and applying multivariate techniques such as SEM could provide a more nuanced understanding of the factors influencing green space accessibility disparities. Future studies should also consider using more sophisticated methods to address multicollinearity and explore specific causal hypotheses.

Moreover, the focus on the MRSP, while relevant, may restrict the generalizability of the findings to other emerging economies with different urban dynamics and socio-political contexts. Comparative studies in other regions of Brazil and the Global South are needed to identify common patterns and challenges, leading to more broadly applicable policy recommendations. Lastly, the study did not include potentially influential variables such as political governance, community engagement, and cultural factors, which could provide a more comprehensive understanding of the drivers of environmental racism in UGS accessibility. Future research should consider these variables to offer a holistic perspective on the issue.

7. Conclusions

This study highlights the crucial influence of socio-economic factors on environmental justice, particularly in UGS accessibility within the MRSP. Findings indicate that while GDP and HDI can exacerbate environmental racism, these impacts vary across populations. A negative correlation between urban vegetation and environmental racism suggests that

urban planning must focus on equitable green space distribution. The interplay between population density and urbanization further complicates equitable UGS allocation, emphasizing the need for innovative planning. Additionally, GDP and HDI's interaction effects highlight the importance of context-specific policies to address disparities. Prioritizing UGS development in affluent areas, enhancing vegetation coverage, and optimizing UGSs in high-density regions are essential for achieving environmental justice. Moreover, incorporating community participation into UGS planning is vital to meet marginalized groups' needs, ensuring more effective and equitable urban green space policies. A nuanced understanding of these dynamics is essential for inclusive urban planning.

Author Contributions: Conceptualization, A.B.; Formal analysis, A.B., A.I.S.L. and R.G.N.; Investigation, A.B. and A.I.S.L.; Methodology, A.B. and A.I.S.L.; Software, R.G.N.; Validation, R.G.N.; Writing—original draft, A.B., A.I.S.L. and R.G.N. All authors have read and agreed to the published version of the manuscript.

Funding: We gratefully acknowledge the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) for its support through the CAPES-Print program (grant number 88887.936587/2024-00), the National Council for Scientific and Technological Development (CNPq) (grant number 401721/2023-0), and the São Paulo Research Foundation (FAPESP) (grant number 2023/03387-5).

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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